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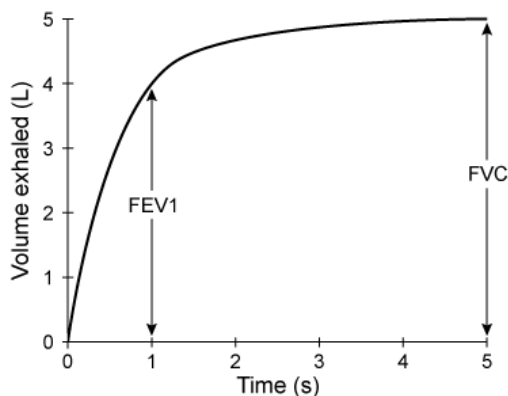


Figure 1 Forced expiratory testing results (spirogram) of a healthy individual

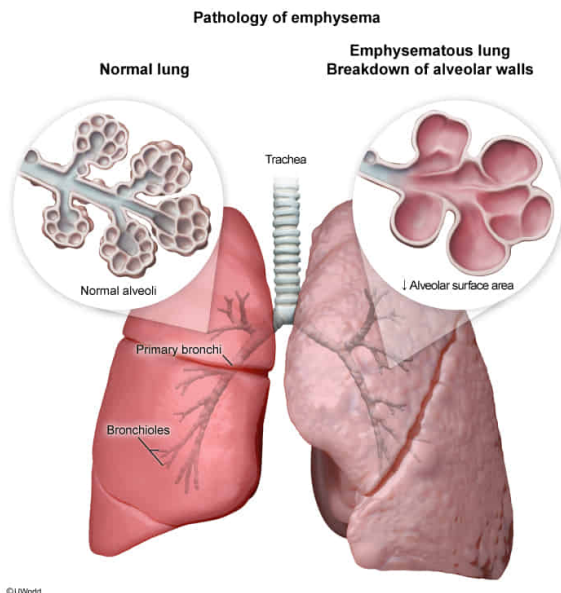
According to the passage, which of the following changes would be expected in a patient with signs of emphysema?

- ☐ A. Narrowing of the trachea [2%]
- ✓ ☒ B. Less efficient gas exchange [80%]
- ☐ C. Weakened diaphragm muscle [7%]
- ☐ D. Overstretched pharyngeal tissues [8%]

Correct

80% answered correctly

Explanation:



The respiratory tract can be divided into the conducting zone and the respiratory zone. The conducting zone includes the nose, pharynx, larynx (voice box), trachea (windpipe), bronchi, and bronchioles. These structures provide passageways for air to travel to and from the respiratory zone, which consists of structures capable of gas exchange.

The majority of gas exchange occurs at the **alveoli**, which are located at the ends of the respiratory tract branches. The thin walls of the alveoli allow oxygen and carbon dioxide to diffuse into the bloodstream (and vice versa) through the pulmonary capillaries surrounding the alveolar sacs. The small size and sheer number of alveoli in the lungs provide an **enormous surface area** available for efficient gas exchange in healthy individuals.

Emphysema is stated in the passage to cause the **breakdown of alveolar walls**, which would decrease total alveolar surface area. Because the alveoli are the main site of gas exchange, **less efficient gas exchange** would be expected because less air can come in contact with respiratory tissue.

(Choice A) The passage states that the narrowing of the airways (eg, trachea) is a characteristic of asthma, not emphysema.

(Choice C) Although emphysema-induced lung hyperinflation can cause the diaphragm (a skeletal muscle) to remain flatter during relaxation, the information in the passage does not suggest that the diaphragm would become weakened.

(Choice D) The **pharynx** is a cavity of the conducting zone located at the back of the mouth. The pharynx is not made up of pulmonary tissues.

Educational objective:

The alveoli are highly efficient at gas exchange due to their structure and extensive surface area. The destruction of alveolar walls in emphysematous lungs is expected to decrease the lungs' ability to perform respiratory gas exchange.

Subject:
Biology

System:
Circulation and Respiration

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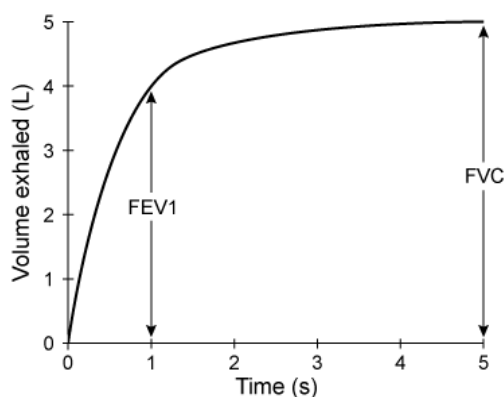
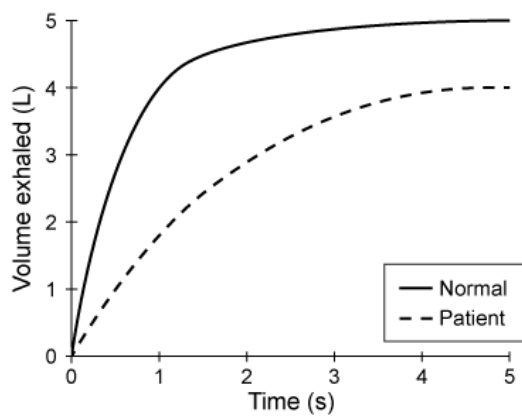


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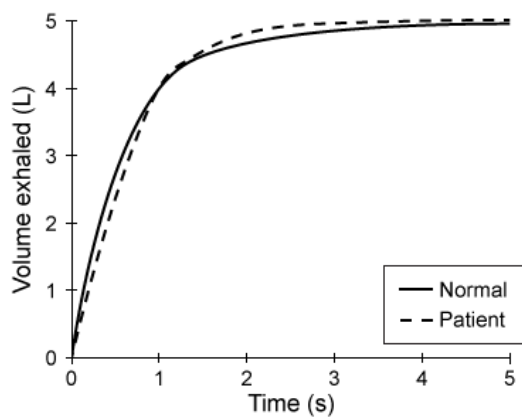
Which of the following FEV1 spiograms would be expected in a patient experiencing symptoms of asthma?

✔ ☒ A.

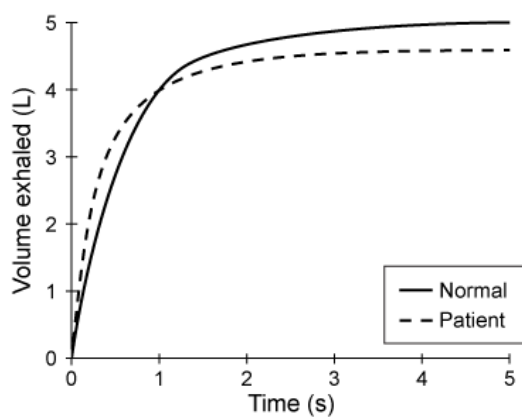


[92%]

☐ B.

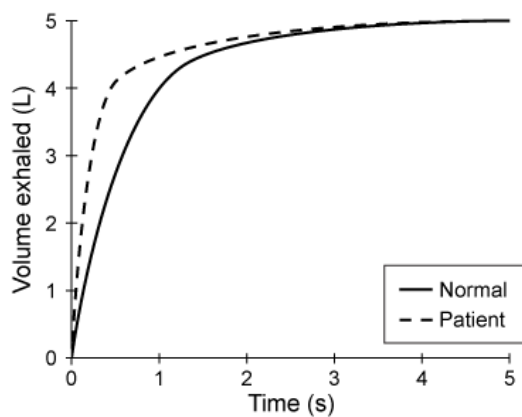


☐ C.



[4%]

☐ D.



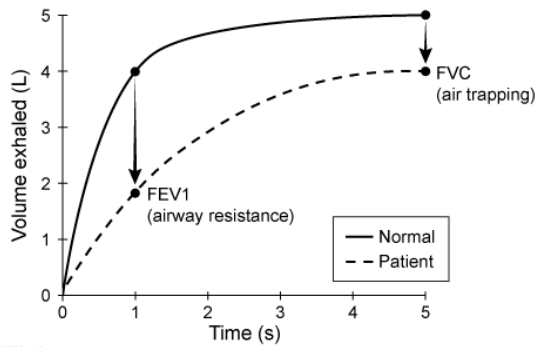
[2%]

Correct

92% answered correctly

Explanation:

Effect of asthma symptoms on spirometry results



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Forced vital capacity (FVC) is defined as the total volume of air exhaled in a single breath. The spirogram of the healthy individual shown in Figure 1 provides baseline values for FVC and the volume of air forcibly exhaled in the first second (FEV1). In a healthy individual, the ratio of FEV1 to FVC is greater than 0.7 and FVC is normal for the person's size.

As stated in the passage, the defining characteristic of patients with COPD is an impaired capacity for exhalation. When experiencing symptoms, a patient with **asthma** exhibits decreased airway diameter and **increased resistance to flow**, resulting in a **decreased volumetric rate** of forcibly **exhaled air**. In addition, the total volume of **exhaled air** (ie, FVC) may be lower if air trapping in the lungs occurs. Therefore, in a patient experiencing asthma symptoms, **FEV1 would decrease** and the **FEV1/FVC** ratio would be **lower** than 0.7.

(Choices B, C, and D) Both FEV1 and FEV1/FVC would be reduced (*not* normal or increased) in a patient experiencing asthma symptoms.

Educational objective:

The narrowing of respiratory airways in asthma increases resistance to airflow, which decreases the volumetric rate of forcibly exhaled air (ie, FEV1 decreases). The narrow airways may also trap air in the lungs, decreasing the total volume of air that can be exhaled (ie, FVC decreases).

Time spent: 0 sec

Subject:
Biology

QID: 400675

System:
Circulation and Respiration

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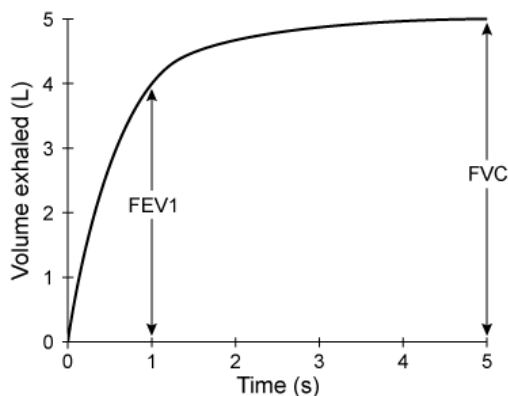


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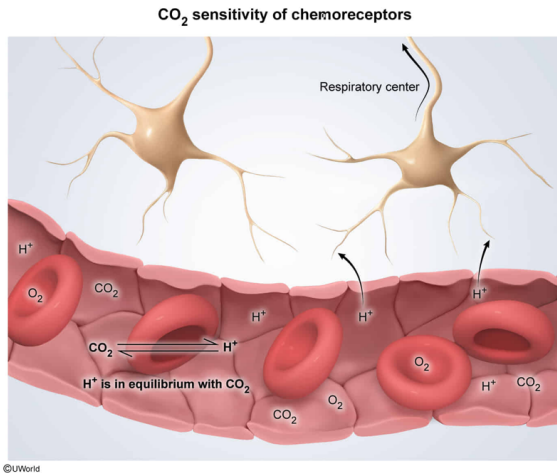
The regulation of respiratory rate is normally most sensitive to:

- ☐ A. PO_2 in the blood. [9%]
- ✓ ☒ B. PCO_2 in the blood. [68%]
- ☐ C. PO_2 in the alveoli. [9%]
- ☐ D. PCO_2 in the alveoli. [11%]

Correct

67% answered correctly

Explanation:



Because it is small and nonpolar, CO_2 in the blood is able to diffuse across membranes and can cross the **blood-brain barrier**. In the brain as well as in the general circulation, **pH** (which is determined by H^+ concentration) is linked to the concentration of CO_2 through the equilibrium of the bicarbonate buffer system:



The equilibrium of this buffer system will shift due to blood PCO_2 changes as well as the resultant effects on brain PCO_2 . **Chemoreceptors** in the brain and periphery indirectly **respond to the PCO_2 in the blood and brain** by directly sensing changes in **$[\text{H}^+]$** .

Information from these receptors is relayed to the respiratory center of the nervous system to regulate respiration.

The regulation of **respiratory rate** is normally most sensitive to the PCO_2 **of the blood**, which is monitored by the central and peripheral chemoreceptors. The central chemoreceptors are located in the brain, and the peripheral chemoreceptors are located in the aortic and carotid arteries.

(Choice A) Although the peripheral chemoreceptors also monitor the partial pressure of O_2 (PO_2) in the blood, PO_2 plays a much less significant role in respiratory regulation than does PCO_2 .

(Choices C and D) The chemoreceptors responsible for the regulation of respiratory rate are located along the major arteries and in the brain, and therefore do not monitor the partial pressures of alveolar gas.

Educational objective:

The regulation of respiratory rate primarily depends on the PCO_2 of the blood as sensed by central and peripheral chemoreceptors. These receptors directly detect the $[\text{H}^+]$ in the blood, which is dependent on the partial pressure of CO_2 in the blood through the bicarbonate buffer system.

Time spent:0 sec
Subject:
Biology

QID:400676
System:
Circulation and Respiration

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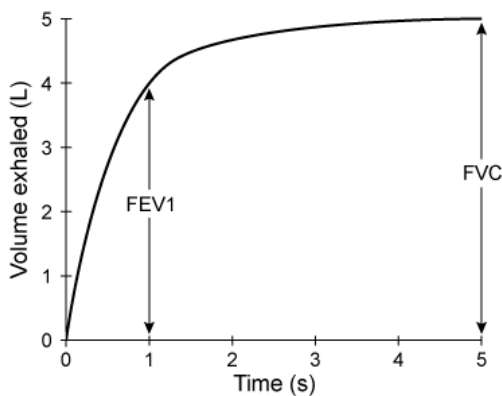


Figure 1 Forced expiratory testing results (spirogram) of a healthy individual

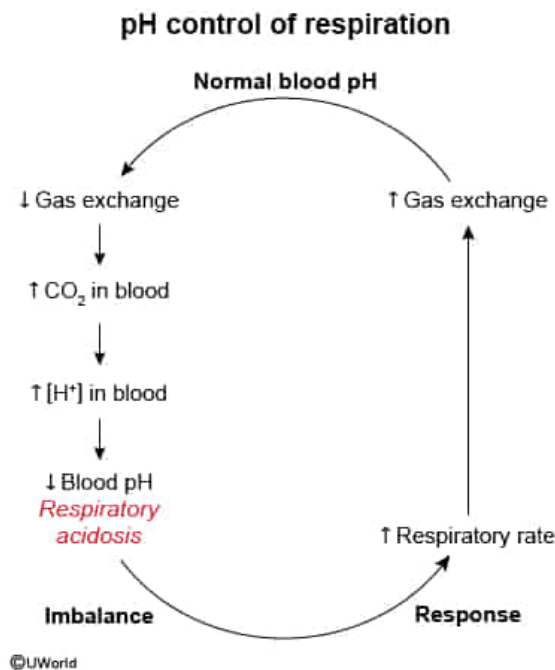
Prolonged bronchoconstriction during an asthma attack would have what effect on blood pH, and what would be the expected homeostatic response?

- ✔ ☒ A. Respiratory acidosis, and an increased respiratory rate [71%]
☐ B. Respiratory acidosis, and a decreased respiratory rate [13%]
☐ C. Respiratory alkalosis, and an increased respiratory rate [10%]
☐ D. Respiratory alkalosis, and a decreased respiratory rate [4%]

Correct

70% answered correctly

Explanation:



One of the major functions of **pulmonary respiration** is the removal of CO₂, a waste product of **cellular respiration**, from the blood through gas exchange at the **alveolar capillary beds**. According to the passage, the constriction of the airways in asthma restricts airflow, and therefore a decreased volume of fresh air is available for gas exchange in the lungs. Due to **less efficient gas exchange**, less CO₂ is removed from the blood and **blood CO₂ will build up**.

Blood pH represents the concentration of H⁺ in the blood and is affected by the partial pressure of CO₂ in the blood through the bicarbonate buffer system:



An increase in blood CO₂ **shifts the equilibrium** to the right, increasing [H⁺] and **decreasing pH**. This decrease in blood pH due to decreased gas exchange is known as **respiratory acidosis**.

In response to the change in blood pH, the respiratory center of the nervous system attempts to **restore the balance** of CO₂ and H⁺ in the blood by changing the rate of respiration. An **increased respiratory rate** (ie, hyperventilation) increases the amount of CO₂ removed from the blood, shifting the equilibrium of the bicarbonate buffer system to the left, decreasing [H⁺] and increasing the blood pH back to homeostatic levels.

(Choice B) A decreased respiratory rate would continue to increase the amount of CO₂ in the blood, further decreasing the pH of the blood (ie, removing the patient further from homeostasis).

(Choices C and D) An increase in blood pH due to respiration is known as **respiratory alkalosis**. Prolonged bronchoconstriction during an asthma attack would lead to a **decrease**

in blood pH (ie, respiratory acidosis).

Educational objective:

Respiratory gas exchange removes CO_2 from the blood, and CO_2 levels affect the pH of the blood by shifting the equilibrium of the bicarbonate buffer system. A decrease in gas exchange causes respiratory acidosis, and the body attempts to restore normal blood pH by increasing the respiratory rate.

Time spent:0 sec

Subject:
Biology

QID:400677

System:
Circulation and Respiration

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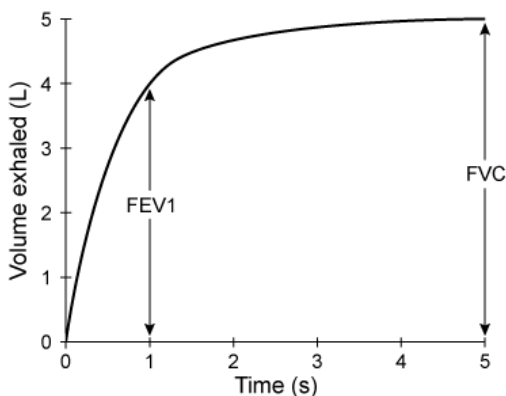


Figure 1 Forced expiratory testing results (spirogram) of a healthy individual

The inflation of the lungs in normal inspiration involves:

- I. Contraction of the diaphragm
- II. Reduction of intrapleural pressure
- III. Elevation of the rib cage

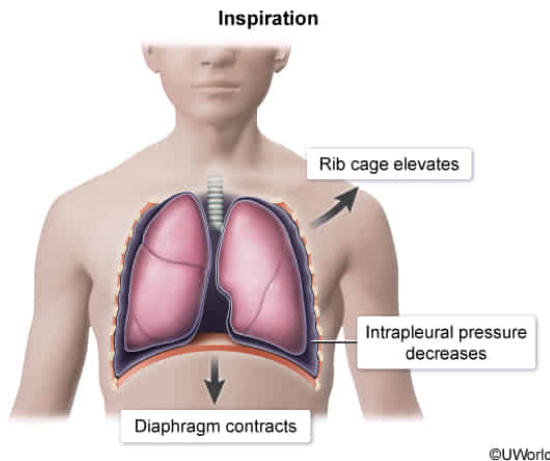
- ☐ A. I and II only [12%]
☐ B. I and III only [20%]
☐ C. II and III only [8%]

✓ ☒ D. I, II, and III [58%]

Correct

58% answered correctly

Explanation:



The lungs are the site of **gas exchange**, increasing in volume as fresh air is drawn in during **inspiration** and shrinking as air exits during expiration. Although they largely fill the thoracic cavity, the lungs are not directly attached to the thoracic walls; instead, they are separated by a thin, fluid-filled space known as the **pleural cavity**.

When the thoracic cavity (chest cavity) expands during inspiration, the **pressure in the intrapleural space decreases** and the **resulting negative pressure** pulls on the lungs, causing them to expand (**Number II**).

Normal (quiet) **inspiration** is primarily initiated by the diaphragm. The diaphragm is a sheet of skeletal muscle below the lungs, and when it is relaxed it curves upward to adopt a dome shape. When the **diaphragm contracts**, it flattens, pulling down on the lungs and causing them to expand (**Number I**). In addition, the external intercostal muscles (a group of muscles at the rib cage) and scalenes (muscles attached to ribs near the neck) can contract to **elevate the rib cage** (thereby preventing the diaphragm from pulling the lower ribs inward) and further expand the lungs (**Number III**).

Educational objective:

The intrapleural space is the thin space between the lungs and the thoracic wall. The contraction of the diaphragm and the elevation of the rib cage decrease the pressure in the intrapleural space, causing the lungs to expand.

Time spent: 0 sec

Subject:
Biology

QID: 400678

System:
Circulation and Respiration

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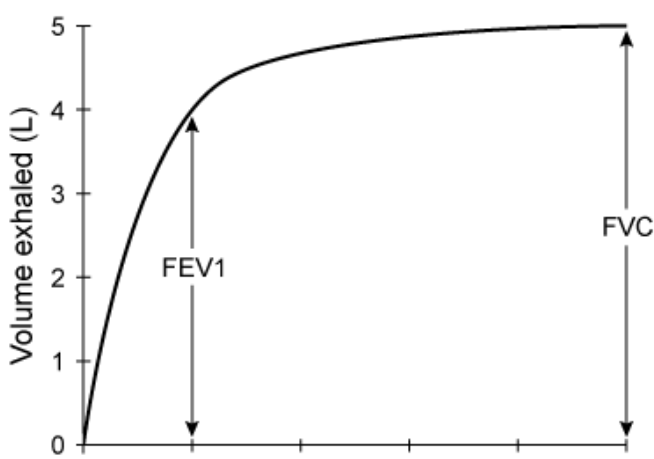


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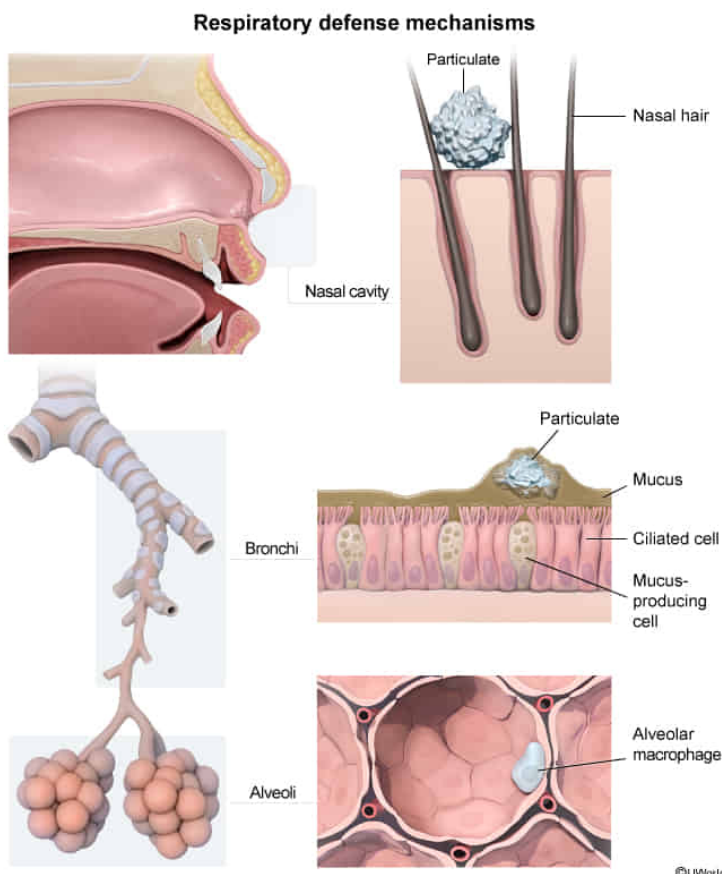
Primary ciliary dyskinesia (PCD), a rare disease that impairs the normal function of cilia, is often misdiagnosed as atypical asthma because wheezing is observed in both illnesses. Unlike asthma, PCD would cause the inability to:

- ☐ A. balance respiratory thermoregulation. [3%]
- ☐ B. cough or sneeze. [4%]
- ✓ ☒ C. remove inhaled particulates. [90%]
- ☐ D. speak in lower pitches.

Correct

89% answered correctly

Explanation:



Cilia are **hair-like organelles** that are found in some specialized cells. Ciliated cells are involved in the defense mechanisms of the respiratory system:

- **Nasal hairs:** Hairs in the nasal cavity can filter out relatively large particulates before they enter the respiratory tract.
- **Mucociliary escalator:** Specialized cells lining the upper respiratory tract produce mucus, a thick and sticky secretion that traps inhaled bacteria and particulates. **Ciliated cells** beat synchronously to **sweep foreign bodies trapped in mucus** up the respiratory tract to the **pharynx**. From the pharynx, these foreign bodies can be coughed out or swallowed and destroyed by the harsh acidity of the stomach.
- **Phagocytosis:** Alveolar macrophages are a type of large white blood cell that engulfs (phagocytizes) and digests foreign material that reaches the alveoli.

Primary ciliary dyskinesia (PCD) is stated to cause **dysfunctional cilia**, which would allow mucus to accumulate in the airways and result in wheezing noises. Subsequently, the **ability to remove inhaled particulates** would be impaired.

(Choice A) Thermoregulation in the respiratory system is achieved through the radiation of heat from nasal and tracheal capillary beds, the convection of hot air throughout the respiratory tract, and the endothermic evaporation of moisture.

(Choice B) Coughing and sneezing are forms of forced expiration that help clear trapped particles from the pharynx.

(Choice D) The larynx (commonly known as the voice box) vibrates at specific frequencies to produce sound. Although cilia are present in the larynx as part of the mucociliary escalator, they do not affect sound production.

Educational objective:

In the upper respiratory tract, mucus-producing cells and ciliated cells work together to trap and remove potentially pathogenic bodies.

Therefore, dysfunctional cilia would decrease the respiratory system's ability to remove inhaled particulates.

Time spent:0 sec

Subject:

Biology

QID:400679

System:

Circulation and Respiration